

INTRODUCTION TO TECHNOLOGY AND LEARNING

- Online Education vs Traditional Education

Online Education - Online education is a form of education which is delivered and administered using the Internet.

Traditional Education - traditional education means education in which instruction takes place between an instructor and students where all are physically present in the same classroom.

UNDERSTANDING THE BASIC CONCEPTS IN ICT

T I T E E A

TECHNOLOGY - the application of scientific knowledge for practical purposes, especially in industry.

INSTRUCTIONAL TECHNOLOGY - refers to the theory and practice of design, development, utilization, management and evaluation of the processes and resources for learning.

TECHNOLOGY IN EDUCATION - Technology in education is also stated as educational technology or Ed Tech. The combination of computer hardware and software with educational theory facilitates the learning of students.

EDUCATIONAL TECHNOLOGY - is the field of study that investigates the process of analyzing, designing, developing, implementing, and evaluating the instructional environment, learning materials, learners, and the learning process to improve teaching and learning.

EDUCATIONAL MEDIA - refers to channels of communication that carry messages with an instructional purpose. They are usually utilized for the sole purpose of learning and teaching.

AUDIO- VISUAL - using both sight and sound, typically in the form of slides or video and recorded speech or music.

- ❖ **Digital Literacy** – ability to use information and communication technologies and the Internet.
- ❖ **Digital Learning** – any type of learning that is accompanied by technology.
- ❖ **Digital tools** – applications. (on-line and offline)
- ❖ **Multimedia**- variety of media formats.

ROLES OF ICT IN TEACHING AND LEARNING according to Stosic (2015)

- **Technology as a tutor** - With technology, tutors and teachers can bring simulations and modeling into the classroom, assisting students who have difficulty grasping the information through verbal discussion.
- **Technology as a teaching tool** - Technology provides students with easy-to-access information, accelerated learning, and fun opportunities to practice what they learn.

- **Technology as a learning tool** - provides students with easy-to-access information, accelerated learning, and fun opportunities to practice what they learn.
- **Technology provides enormous support to the teacher** - Technology provides enormous support to the teacher as the facilitator of learning. It transforms a passive classroom to an active and interactive one, with audio-visual aids and models, smart classrooms, e-learning classrooms which motivate and increase attention level of learners.
- **Technology has modernized the teaching-learning environment**- Technology provides students with easy-to-access information, accelerated learning, and fun opportunities to practice what they learn. It enables students to explore new subjects and deepen their understanding of difficult concepts, particularly in STEM.
- **Technology opens new fields in educational researches** - Through technology, researchers can process data faster and more accurate.
- **Technology supports teacher professional development** - Professional development utilizing technology should involve learning content in context and modeling pedagogically appropriate methods.
- **Support learners to learn how to learn on their own**
- **Enhances learners' communication skills through social interactions.**
- **Upgrades learners' HOTS: critical thinking, problem solving and creativity.**

ICT AND ASSESSMENT

Current Trends In Assessment Of Learning

- ✓ Multiple measures
- ✓ Increasing the use of technology
- ✓ Focus on growth of the learner
- ✓ Differentiating the roles and responsibilities

CAA (Computer-Assisted Assessment)

-any form of assessment delivered through computers.

Computer-assisted assessment is used to assess learning and performance of the students. It includes all types of assessments whether formative or summative assessment and which is then conveyed with the assistance of these modern technologies. Delivery of the assessment is made online as well as offline.

E-Portfolios: -a digital compilation of learning records of a student.

Electronic portfolio is a process of gathering of e - evidence collected and maintained by a person on the web. Such kind of e-evidence include written text, files, multimedia, images and hyperlinks. The E- portfolios is basically useful for assessing the abilities of the person.

Digital Rubrics: -includes the descriptions in the subjective manner about the performance of a student.

After making the use of a rubric, the student can access their weakness. The way of assessment in rubrics follows the formative type of assessment as student can know the weakness in any area where he/she needs to improve. Teachers in the schools can extend their support to students in the construction of a rubric.

Survey Tools: -used to collect feedback from students regarding any manner.

ICT POLICIES, ISSUES, AND GUIDELINES

ICT policies are needed to put a roadmap or course of actions to be pursued and adopted by various governments, org, entities, involving ICT

SOME ISSUES ON ICT

- ISSUES NO. 1: Freedom of Expression and Censorship
- ISSUES NO. 2: Privacy and Security
- ISSUES NO. 3: Surveillance and Data Retention
- ISSUES NO. 4: E-pollutants from E-waste

E-SAFETY

- Helps safeguard children and young people in the digital world
- Emphasizes learning to understand and new technologies in a positive way
- Educates children about the risks as well as the benefits so we can feel confident online.
- Supports young learners and adults to develop safer online behaviors, both in and out of school

NETWORK MANAGEMENT

1. Safety in the Use of Network in Schools
2. Password Policy
3. Personal mobile phones and mobile devices
4. Cameras

DIGITAL LITERACY SKILLS IN THE 21ST CENTURY

Digital literacies are the individual's capabilities to be able to effectively and responsibly function and perform in digital society. The term "**digital literacy**" was coined by **Paul Gilster in 1997**.

THE DIGITAL LITERACIES

MEDIA LITERACY – one's ability to critically read information and content.

INFORMATION LITERACY – locating information from the web and interpreting and evaluating its validity.

ICT LITERACY – knowing how to select and use digital devices, apps, and services.

COMMUNICATION & COLLABORATION – one's capabilities in being able to participate in digital networks.

Online Communities/Social Networking

1. Facebook

Facebook is an online social media and social networking service owned by American company Meta Platforms. Founded in 2004 by Mark Zuckerberg with fellow Harvard College students and roommates Eduardo Saverin, Andrew McCollum, Dustin Moskovitz, and Chris Hughes, its name comes from the face book directories often given to American university students. As of July 2022, Facebook claimed 2.93 billion monthly

active users, and ranked third worldwide among the most visited websites as of July 2022. It was the most downloaded mobile app of the 2010s.

2. Twitter

Twitter is a microblogging and social networking service owned by American company Twitter, Inc., on which users post and interact with messages known as "tweets".

Twitter was created by Jack Dorsey, Noah Glass, Biz Stone, and Evan Williams in March 2006 and launched in July of that year.

3. LinkedIn

LinkedIn is an American business and employment-oriented online service that operates via websites and mobile apps. Launched on May 5, 2003, the platform is primarily used for professional networking and career development, and allows job seekers to post their CVs and employers to post jobs. From 2015 most of the company's revenue came from selling access to information about its members to recruiters and sales professionals. Since December 2016, it has been a wholly owned subsidiary of Microsoft. As of February 2022, LinkedIn has 830+ million registered members from over 200 countries and territories.

4. Google

Google was founded on September 4, 1998, by Larry Page and Sergey Brin while they were PhD students at Stanford University in California. Google is the "most powerful company in the world" and one of the world's most valuable brands due to its market dominance, data collection, and technological advantages in the area of artificial intelligence.

5. Youtube

YouTube is an American online video sharing and social media platform headquartered in San Bruno, California. It was launched on February 14, 2005, by Steve Chen, Chad Hurley, and Jawed Karim. It is owned by Google, and is the second most visited website, after Google Search.

6. Pinterest

Pinterest is an image sharing and social media service designed to enable saving and discovery of information (specifically "ideas") on the internet using images, and on a smaller scale, animated GIFs and videos, in the form of pinboards. The site was created by Ben Silbermann, Paul Sciarra, and Evan Sharp, and had 433 million global monthly active users as of July 2022. It is operated by Pinterest, Inc., based in San Francisco.

7. Instagram

Instagram is a photo and video sharing social networking service owned by American company Meta Platforms. The app allows users to upload media that can be edited with filters and organized by hashtags and geographical tagging. Posts can be shared publicly or with preapproved followers. Users can browse other users' content by tag and location, view trending content, like photos, and follow other users to add their content to a personal feed.

8. Tumblr

Is a (stylized as tumblr; pronounced "tumbler") is an American microblogging and social networking website founded by David Karp in 2007 and currently owned by Automattic.

The service allows users to post multimedia and other content to a short-form blog. Users can follow other users' blogs. Bloggers can also make their blogs private. For bloggers, many of the website's features are accessed from a "dashboard" interface. As of July 2021, Tumblr hosts more than 529 million blogs.

9. Flickr

Flickr is an American image hosting and video hosting service, as well as an online community, founded in Canada and headquartered in the United States. It was created by Ludicorp in 2004 and was a popular way for amateur and professional photographers to host high-resolution photos. It has changed ownership several times and has been owned by SmugMug since April 20, 2018.

10. Bebo

Bebo is an American social networking website that originally operated from 2005 until its bankruptcy in 2013 and relaunched in February 2021. The site relaunched several times after its bankruptcy with a number of short-lived offerings, including instant messaging and video streaming, until its acquisition by Amazon in July 2019 when it was shut down. It was announced in January 2021 that it would be returning as a new social media site the month after. By May 2022, it had been once again shutdown, without having ever left beta testing.

Benefits And Dangers In Using Social Networking Sites

Benefits

1. Technology Literacy
2. Breadth of Knowledge
3. Bringing people together
4. Attracts attention
5. Opportunity to widen business

Dangers

1. Distraction
2. Relationship problems
3. Health problems
4. Illegal crimes and virus attacks
5. Negatively affecting worker productivity

IDENTITY MANAGEMENT – understand how to ensure safety and security in managing online identity.

Safeguarding oneself in using social networking sites

1. Secure password
2. Log-out every time
3. Digital footprint
4. Do not accept invites from strangers
5. Regularly check privacy settings
6. Seriously consider the effect of possible posts on others
7. Using appropriate language
8. Better be safe than sorry
9. Report button
10. Cyberbullying

LEARNING SKILLS – knowing how to study and learn in a technology-enriched environment.

DIGITAL SCHOLARSHIP– think and participate in professional and research practices.

8 DIGITAL LITERACY SKILLS (LYNCH, 2017)

CODING - the process of assigning a code to something for classification or identification.

COLLABORATION - the action of working with someone to produce or create something.

CLOUD SOFTWARE - is application software that is hosted in the cloud and that you access and use via a web browser, a dedicated desktop client, or an API that integrates with your desktop or mobile operating system.

WORD PROCESSING - refers to the act of using a computer to create, edit, save, and print documents. To perform word processing, specialized software (known as a Word Processor) is needed.

SCREEN CASTING - is a digital video and audio recording of what occurs on a presenter's computer screen, and it can be used to create sophisticated, information-rich multimedia presentations.

ARCHIVING - a place in which public records or historical materials (such as documents) are preserved an archive of historical manuscripts a film archive.

INFORMATION EVALUATION - encourages you to think critically about the reliability, validity, accuracy, authority, timeliness, point of view or bias of information sources.

SOCIAL MEDIA - is a collective term for websites and applications that focus on communication, community-based input, interaction, content-sharing, and collaboration.

RA 8293

AN ACT PRESCRIBING THE INTELLECTUAL PROPERTY CODE AND ESTABLISHING
THE INTELLECTUALPROPERTY OFFICE, PROVIDING FOR ITS POWERS AND
FUNCTIONS, AND FOR OTHER PURPOSES

ICT Competency Standards for Pre-Service Teacher

- **Domain 1: Understanding ICT in Education**
 - 1.1. Demonstrate awareness of policies affecting ICT in education
 - 1.2. Comply with ICT policies as they affect teaching-learning environment
 - 1.3. Contextualize ICT policies to the learning environment
- **Domain 2: Curriculum and Assessment**

- 2.1. Demonstrate understanding of concepts, principles and theories of ICT systems as they apply to teaching-learning
- 2.2. Evaluate digital and non-digital learning resources in response to student's diverse needs.
- 2.3. Develop digital learning resources to enhance teaching-learning
- 2.4. Use ICT tools to develop 21st century skills: information media and technology skills, learning skills, learning and innovation skills, career skills and effective communication skills.

- **Domain 3: Pedagogy**

- 3.1. Apply relevant technology tools for classroom activities
- 3.2. Use ICT knowledge to solve complex problems and support students' collaborative activities
- 3.3. Model collaborative knowledge construction in face to face and virtual environment

- **Domain 4: Technology Tools**

- 4.1 Demonstrate competence in the technical operations of technology tools and systems as they apply to teaching and learning
- 4.2 Use technology tools to create new learning opportunities to support community of learners
- 4.3 Demonstrate proficiency in the use of technology tools to support teaching and learning

- **Domain 5: Organization and Administration**

- 5.1 Manage technology-assisted instruction in an inclusive classroom environment
- 5.2 Exhibit leadership in shared decision-making using technology tools

- **Domain 6: Teacher Professional Learning**

- 6.1. Explore existing and emerging technology to acquire additional content and pedagogical knowledge
- 6.2 Utilize technology tools in creating communities of practice
- 6.3 Collaborative with peers, colleagues, and stakeholders to access information in support of professional learning

- **Domain 7: Teacher Disposition**

- 7.1 Demonstrate social, ethical, and legal responsibility in the use of technology tools and resources
- 7.2 Show positive attitude towards the use of technology tools

ISTE Standards for Students and Teachers (Philippine Setup)

ISTE Standards for Students

1. **Creativity and Innovation** – students express themselves creatively for a variety of purposes using platforms, tools, styles, and formats appropriate for their goals.
2. **Communication and Collaboration** – students use digital tools to connect with learners from a variety of backgrounds and cultures in ways that broaden mutual understanding and learning.

3. **Research and Information Fluency** - Students plan and employ effective research strategies to locate information and other resources for their intellectual or creative pursuits.
4. **Critical Thinking, Problem-Solving & Decision Making** - students develop and employ strategies for understanding and solving problems in ways that leverage the power of technological methods to develop and test solutions.
5. **Digital Citizenship** - students recognize the rights, responsibilities, and opportunities of living, learning, and working in an interconnected digital world, and they act and model in ways that are safe, legal, and ethical.
6. **Technology Operations and Concepts** - students leverage technology to take an active role in choosing, achieving, and demonstrating competency in their learning goals, informed by the learning sciences.

ISTE Standards for Teachers

1. **Technology Operations and Concepts** - educators continually improve their practice by learning from and with others and exploring proven and promising practices that leverage technology to improve student learning.
2. **Planning and Designing Learning Environment and Experiences** - educators design authentic, learner-driven activities and environments that recognize and accommodate learner variability.
3. **Teaching, Learning and Curriculum** - educators facilitate learning with technology to support student achievement of the ISTE Standards for Students. Aligned with the curriculum
4. **Assessment and Evaluation** - educators understand and use data to drive their instruction and support students in achieving their learning goals. Provide alternative ways for students to demonstrate competency and reflect on their learning using technology.
5. **Productivity and Professional Practice** - model for colleagues the identification, exploration, evaluation, curation and adoption of new digital resources and tools for learning.
6. **Social, Ethical, Legal and Human Issues** - educators inspire students to positively contribute to and responsibly participate in the digital world. Create experiences for learners to make positive, socially responsible contributions and exhibit empathetic behavior online that build relationships and community.

Promote OER Creation – Having the technology to access learning resources. Engaging educational content must be produced, updated, repurposed, and distributed regularly.

Appropriate Technologies – The choice of technology should be practical, cost-effective, and sustainable. Use technologies that are accessible to all learners including TV, radio, and print.

Donor Coordination and Harmonization – Initiatives and beneficiaries should be well coordinated. Connecting projects with the most appropriate recipients helps lessen the cost and maximize impact.

Public Private Sector Financing – The private sector must be motivated to participate. Investing in ICT4E will help ensure the continuity of the computer-literate workforce.

THEORIES AND PRINCIPLES IN THE USE OF TECHNOLOGY-DRIVEN LESSONS

ADDIE Model

Analysis

The Analysis phase can be considered as the “Goal-Setting Stage.” The focus of the designer in this phase is on the target audience. It is also here that the program matches the level of skill and intelligence that each student/participant shows.

Design

This stage determines all goals, tools to be used to gauge performance, various tests, subject matter analysis, planning and resources. In the design phase, the focus is on learning objectives, content, subject matter analysis, exercise, lesson planning, assessment instruments used and media selection.

Development

The Development stage starts the production and testing of the methodology being used in the project. In this stage, designers make use of the data collected from the two previous stages and use this information to create a program that will relay what needs to be taught to participants. If the two previous stages required planning and brainstorming, the Development stage is all about putting it into action.

Implementation

The implementation stage reflects the continuous modification of the program to make sure maximum efficiency and positive results are obtained. Here is where IDs strive to redesign, update, and edit the course in order to ensure that it can be delivered effectively.

Evaluation

The last stage of the ADDIE method is Evaluation. This is the stage in which the project is being subjected to meticulous final testing regarding the what, how, why, when of the things that were accomplished (or not accomplished) of the entire project. This phase can be broken down into two parts: Formative and Summative.

ASSURE Model

Analyze Learners

The first step in the process is that the teacher should analyze the attributes of her learners. There should be a focus on those learner characteristics which are associated with the learning outcomes desired. The information gathered will help you in the decisions that you make with respect to the other steps in the process.

State Standards and Objectives

After the analysis of the learner attributes, the teacher must state standards and objectives for the learning module. This statement consists of a specification of what the learners will be able to do as a result of the instruction.

Select Strategies, Technology, Media, and Materials

The second "s" in the acronym stands for select strategies, technology, media, and materials. Given what your learning objectives are, it is necessary to pick instructional strategies, technology, and media that will bring about the results that you want.

Utilize Technology, Media, and Materials

This step in the ASSURE process concerns making a plan as to how you will utilize the technology, media, and materials that you have selected. As with all of the instructional steps, you must make sure that your plans contribute towards producing the objectives that you have laid down.

Require Learner Participation

This step actually belongs within earlier steps. It requires that you make plans for how you are going to actively engage your students in the material that you are teaching. This needs to be figured out both at the class level and the individual level.

Evaluate and Revise

The last step in the ASSURE process is just as crucial as all of the others. In this step, you evaluate the impact of your teaching on student learning. This includes an evaluation of your teaching strategies and the technology, media, and materials that you used.

SAMR (Substitution, Augmentation, Modification, Redefinition) Model

Substitution

At this stage, technology is directly substituted for a more traditional teaching tool or method. It is a simple, bare-bones, direct replacement.

Augmentation

The technology is again directly substituted for a traditional tool or method, but with significant enhancements to the student experience. Ask yourself if the technology increases or augments a student's productivity and potential in some way.

Modification

In this stage, you are beginning to move from enhancement to transformation using the SAMR Model. Instead of replacement or enhancement, this is an actual change to the lesson's design and its learning outcome.

Redefinition

The last stage of the SAMR model represents the pinnacle of how integrated classroom technology can transform a student's experience. In this case, you ask yourself if the technology tools allow educators to redefine a traditional learning task in a way that would not be possible without the tech, creating a novel experience.

TPACK (Technology, Pedagogy and Content Knowledge) Framework

Content Knowledge (CK) – This describes teachers' own knowledge of the subject matter.

Pedagogical Knowledge (PK) – This describes teachers' knowledge of the practices, processes, and methods regarding teaching and learning.

Technological Knowledge (TK) – This describes teachers' knowledge of, and ability to use, various technologies, technological tools, and associated resources.

TECHNOLOGY-ENHANCED LESSONS

Non-Digital And Digital Skills And Tools In Delivering Technology-Enhanced Lessons

Writing Board - work surface consisting of a wide lightweight board that can be placed across the lap and used for writing.

Diorama - a model representing a scene with three-dimensional figures, either in miniature or as a large-scale museum exhibit.

Nature Table - A table on which items collected from nature, such as seeds, rocks, birds' nests, etc., are displayed to teach children about the natural world.

Flip Chart- a large pad of paper bound so that each page can be turned over at the top to reveal the next, used on a stand at presentations.

Zigzag Board - It is a multi-board series of three or four rectangular boards. They are joined together along the sides by hinges so that they can be easily folded up and carried.

Board/Wall Display- students use display boards to promote clubs, recognize athletics and display art projects. The purpose of the display board in that context is to catch the viewer's attention and explain what was performed and what was learned.

Role And Pole Display- Display boards help children's education by assisting with visual learning; teachers use display boards to highlight images relevant to the lesson subject.

Guidelines in the Construction and Development of Conventional Instructional Support Materials:

1. **Unity** - the quality or state of being one
2. **Simplicity** - the quality or condition of being easy to understand or do.
3. **Legibility** - the quality of being clear enough to read.

4. **Consistency** - always behaves in the same way, has the same attitudes towards people or things, or achieves the same level of success in something
5. **Clarity** - the quality of being coherent and intelligible.

Digital Tools Or Non-Conventional Instructional Support For Teaching And Learning.

Mobile Phones And QR Code - A QR code is a type of barcode that can be read easily by a digital device and which stores information as a series of pixels in a square-shaped grid.

Infographics - a visual image such as a chart or diagram used to represent information or data.

Online Bulletin Board - is a qualitative approach that gathers a virtual assembly of people to facilitate data production through interactive discussion among participants and a moderator.

E-Portfolio - is a collection of work (evidence) in an electronic format that displays learning over time.

COLLABORATIVE ENVIRONMENT IN THE DIGITAL WORLD

SKYPE - is software that enables the world's conversations. Millions of individuals and businesses use Skype to make free videos and voice one-to-one and group calls, send instant messages and share files with other people on Skype.

WIKI - a website that allows collaborative editing of its content and structure by its users.

BLOGGING - is the process of writing a blog, an online journal in which you share your thoughts about a particular subject with readers.

GOOGLE GROUP/FORM - Organizations, classes, teams, and other groups can use Google Groups to do things such as: Find people with similar hobbies or interests and take part in online conversations.

Multimedia Learning

Reducing Extraneous Processing

1. **Coherence** - Delete irrelevant material
2. **Signaling** - Highlight essential content
3. **Redundancy** - Do not add captions to narrated graphics
4. **Spatial contiguity** - Place words near corresponding graphics
5. **Temporal Contiguity** - Present spoken words at the same time as graphics

Managing Essential Processing

1. **Segmenting** - Breaking video lessons into learner-paced parts.
2. **Pre-Training** - Present key concepts before the actual lesson.
3. **Modality** - use of spoken words rather than printed words

Fostering Generative Processing

1. **Personalization** – Put words in conversational style rather than formal style.
2. **Voice** – use actual human voice for narrations instead of automated voice
3. **Embodiment** – video presenters must have human-like gestures.

FLEXIBLE LEARNING ENVIRONMENT

Synchronous

- Students learn at the same time
- Communication happens in real time
- Instant feedback

Asynchronous

- Students learn at various times
- Not live
- Students work at their own pace

Blended Learning

a style of education in which students learn via electronic and online media as well as traditional face-to-face teaching.

Models of Blended Learning

Face-to-Face Driver

As the name implies, the face-to-face model of blended learning involves a great deal of student-teacher interaction. It is primarily (although not always) delivered through classroom teaching, where students are in physical attendance.

Lab Rotation Blended Learning Model

This is another model of blended learning which works by allowing students to rotate through digital stations on a fixed schedule, i.e., learners are required to physically change rooms. Learners rotate to a computer lab for online learning activities rather than staying within the same classroom.

The 'Flipped Classroom' Blended Learning Model

The most widely known version of blended learning, a 'Flipped Classroom' is where students are introduced to the content at home. They practice working through it at school supported by a teacher and/or peers. In this method, the traditional role of each other is flipped. Such as students watching online lectures and then collaborating in online discussions. This also means that the students carry out research at home while actively engaging in concepts in the classroom.

Flex Blended Learning Model

In the Flex model, most of the curriculum is delivered via a digital platform. Here, online learning is the backbone of student learning and teachers are available for face-to-face consultation and support. The students work independently and learn to develop/create new concepts in a digital environment.

À La Carte Blended Learning Model

In the À La Carte blended learning model, learners take an online course in addition to the core curriculum. This means that students have the option to choose courses online

to supplement their existing course load. The students are expected to get the work done by a given due date.

Online Driver Blended Learning Model

Opposite to the face-to-face model, the online driver model is where students work remotely, and the material is delivered via an online platform. Although face-to-face check-ins are optional, students can usually chat with teachers online if necessary. This model of blended learning is perfect for students who need more flexibility and independence in their daily schedules.

Online Lab Blended Learning Model

As schools increasingly face tighter resource constraints, the online lab model is a viable option for helping students complete their courses including those not offered at the specific site. In this scenario, all of the curricula are delivered via a digital platform but in a consistent physical location (such as a computer lab). Individuals supervise the lab, but they are not teachers who are trained.

Self-Blend Blended Learning Model

The self-blend model of blended learning allows students to augment their traditional learning with online coursework. For this model of blended learning to be successful, students need to be highly motivated. It is ideal for students who want to take additional, advanced placement courses or those who are interested in a subject area that is not covered in the traditional course catalog.

LEARNING MANAGEMENT SYSTEMS

- deploy and track **online training** initiatives
- develop** online **training materials**
- store and track** information

LMS (Learning Management Systems) Options

Cloud-Based

- Learning is done **online**
- Needs **log-in** access
- No installation** needed
- Requires a **browser** to access

Desktop Application

- The LMS app is **installed** on the **desktop computer**.
- May be accessed **online** or **offline** depending on subscription.
- Online access is **required** during submission

Mobile Application

- accessible **whenever, wherever** via mobile devices.
- may only access **limited features** due to screen size and memory

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